

AMOGH SOMISETTY

(925) 236-2600 · somisett@calpoly.edu · amogh.site · github.com/Asomisetty27

Etched
Hiring Team, Firmware Internship

Dear Etched Team,

I am a third-year Electrical Engineering student at Cal Poly applying for the Firmware Intern role. I build hardware from the register level up, and I spend the rest of my time on the reliability and efficiency of the AI-inference infrastructure your chips are built to replace. That combination is uncommon for an intern, and it is exactly the intersection Etched works at.

On the hardware side: I designed an RV32I CPU from the micro-architecture up in SystemVerilog and brought it up on an FPGA through the full synthesis, place-and-route, and timing-closure flow. I write register-level firmware on AVR and STM32, and I own the avionics electrical design for Cal Poly's autonomous drone team, including a DroneCAN and high-speed-serial interconnect board I verified against the flight-controller pinout in KiCad. I am comfortable living at the boundary between silicon, firmware, and the board it runs on.

On the AI-infrastructure side: I founded Theta, an open-source tool that computes per-GPU effective thermal resistance from live telemetry to catch degraded accelerators that standard monitoring misses. I validated it blind on a production H100 fleet at a top research university, and the operator's own maintenance records independently confirmed the units I flagged. That work taught me, concretely, where GPU inference is inefficient and where it fails, which is the problem Etched is solving in silicon with Sohu.

What I would bring is careful, verification-first engineering (I hold myself to checking every result against a null before I trust it) and a genuine understanding of the transformer-inference workloads your hardware targets. I would be glad to help bring up firmware on chips built to run those models an order of magnitude faster.

Thank you for your time and consideration.

Sincerely,
Amogh Somisetty